

Elevator 4.0: Smarter, Safer, Faster

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Feature 1: Door Safety Pads

Problem: People can get caught in closing elevator doors, risking injury



Solution: Install rubber or soft padding along the edges of both doors

Pros: Protect injuries

Cons: Cost

Feature 2: Cancel Request

- **Problem:** Accidental button presses (up/down) cause unnecessary stops.
- **Solution 1:** Add a dedicated Cancel Button
 - Pros: Simple, intuitive for users
 - Cons: Hardware cost
- **Solution 2:** Implement ***Click Pattern Recognition***
 - Pros: No extra hardware
 - Cons: Less user-friendly; requires training or instructions; software cost
 - Example patterns:
 - Rare manual sequences (e.g., press 5 times consecutively)
 - Machine Learning detects unusual user behavior over time
- Avoid conflicts with normal pressing habits (e.g., double/triple presses to hurry, long holds for door reopen)
- Cancel pattern must be rare and intentional

Feature 2: Cancel Request with ML

Collect Button Press Data for each press event:

- Button type (up/down)
- Timestamp
- Duration of hold
- Number of consecutive presses

Feature Engineering:

- Average interval between presses
- Average duration of button holds
- Pressed sequences: [press, press, press], [press, press], [press, hold, press], ...

Feature 2: Cancel Request with ML

Use unsupervised learning

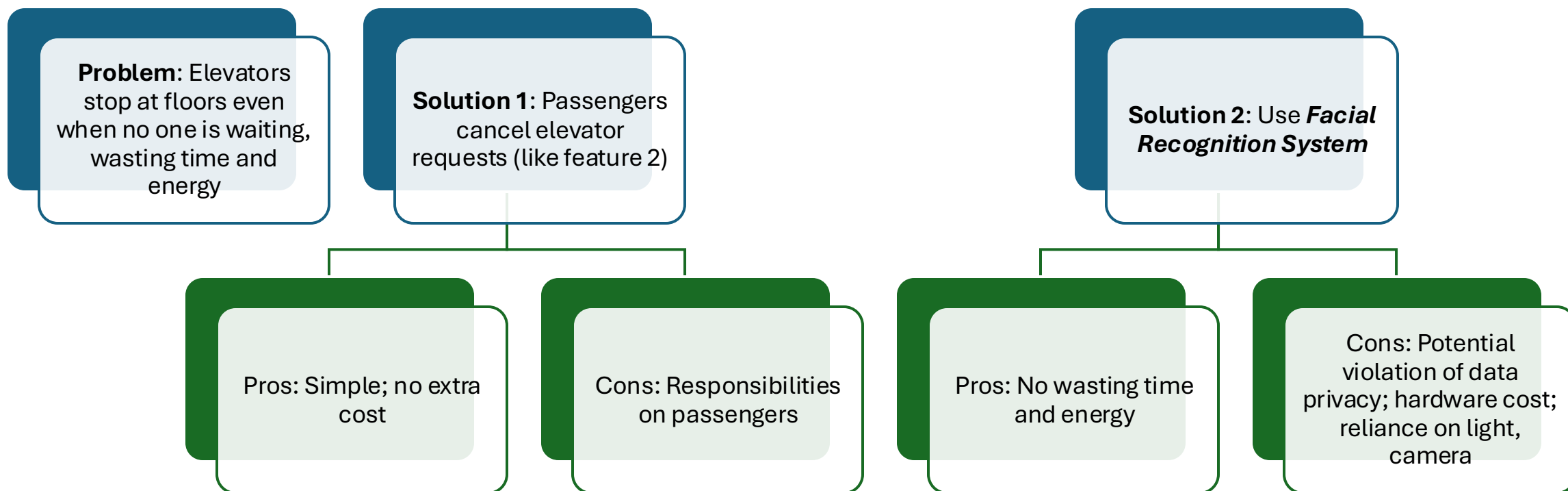
- No labeled data for cancel patterns
- Cluster normal usage patterns
- Seeking anomaly (abnormal patterns)

Detect unusual behavior

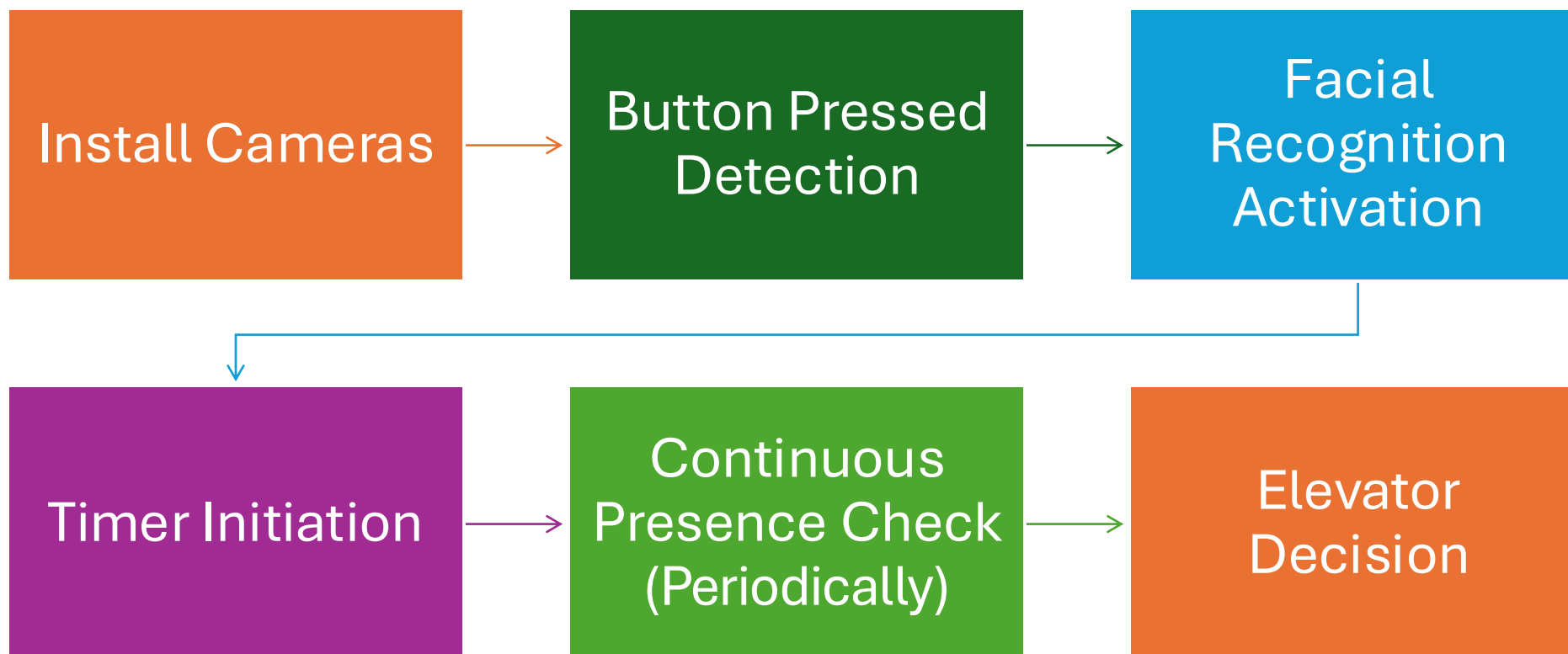
- Ignores patterns that match normal usage to prevent false positives
- Flags sequences that are unusual in real time

Optional Feedback Loop: RL with human evaluation

Feature 3: Detect Waiting Passengers



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Feature 4: Traffic Optimization with ML

Problem: Elevators waste time and get crowded because we don't know which floors people will go to



Solution: ML + Optimization

Pros: Improves elevator efficiency; reduces wait times

Cons: Unpredictable behaviors

Feature 4: Traffic Optimization with ML

- Collect Data Inside the Elevator for every ride:
 - Origin floor → destination floor
 - Timestamp (hour, day of week)
 - Number of passengers
 - Elevator direction (up/down)
- Feature Engineering:
 - Group trips by time windows (e.g., morning, lunch, evening)
 - To find patterns:
 - Floors with frequent boarding at certain times
 - Floors where people often get off at certain times

Feature 4: Traffic Optimization with ML

Clustering using Heatmaps:

- Cluster floors by traffic patterns per time window
- Example: Floors 1–3 busy 8–9am, Floor 10 busy 5–6pm

Predict future traffic with ML (time series or regression)

Scheduling optimization

- Use learned patterns to pre-position elevators
- Send elevators to high-demand floors in advance during peak times
- Reduce stops at low-demand floors if no call is expected
- Combine with real-time button presses (e.g., someone pressing unexpectedly)

Optional Feedback Loop: RL with human evaluation

Thank you :)

Questions and discussions are always
welcome!